



RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES

B.Sc. in Applied Sciences

Second Year Semester I Examination – August / September 2024

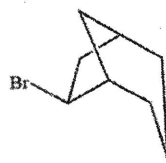
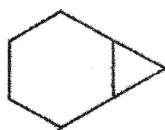
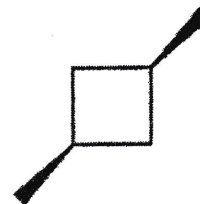
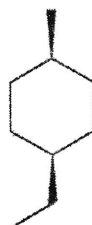
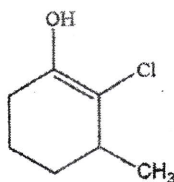
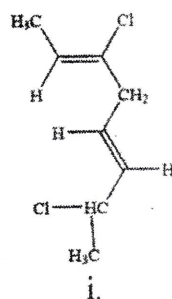
CHE 2202 – ORGANIC CHEMISTRY II

Answer All Questions.

Time: Two (02) hours.

1)

- a) Name the following compounds using R/S, E/Z or cis/trans nomenclature. Draw necessary steps and write IUPAC names of the compounds.



(09 marks)

- b) Draw the Fischer projection of,

- (2R,3S)-3-bromo-2-butanol
- (2R,3R)-3-bromo-2-butanol

(08 marks)

- c) Draw the conformations of Butane using Newman projection formulae and plot the potential energy Vs. angle of rotation curve for the rotation of C2 – C3 bond through 360°C.

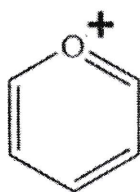
(08 marks)

2)

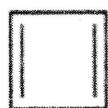
- a) "The addition of H_2 to $\text{C}=\text{C}$ normally gives off about 118 kJ mol^{-1} and benzene has three double bonds would give off only 206 kJ mol^{-1} on reacting with three H_2 molecules". Explain this statement.

(04 marks)

- b) Explain the aromaticity of the following compounds using Huckel's rule.



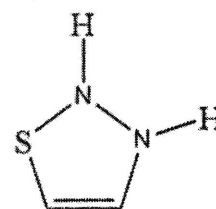
i.



ii.



iii.



iv.

(06 marks)

- c) State and briefly describe 02 limitations of Friedle Craft Alkylation using necessary chemical reactions.

(06 marks)

- d) "The aromatic sulfonation which has been called as one of the most important reactions in industrial organic chemistry". Figure out the mechanism of sulfonation of benzene. State 03 uses of benzene sulfonic acid.

(05 marks)

- e) Outline the reaction mechanism of Friedle craft acylation of benzene.

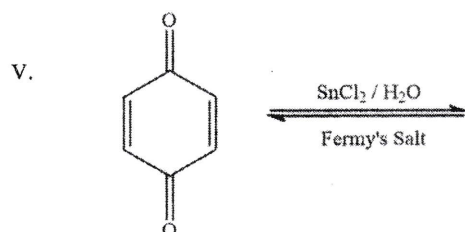
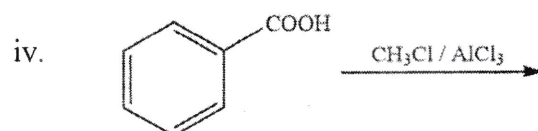
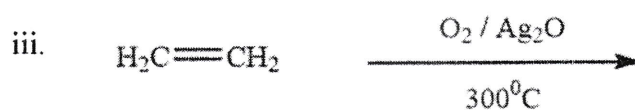
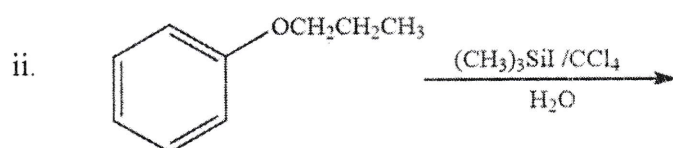
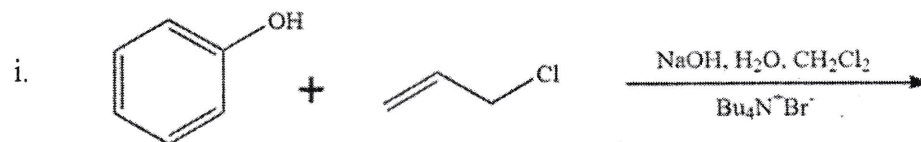
(04 marks)

3)

- a) "Substituted phenols can be more or less acidic than phenol itself". Compare the acidities of meta-nitrophenol and para-nitrophenol by examining their reaction mechanisms.

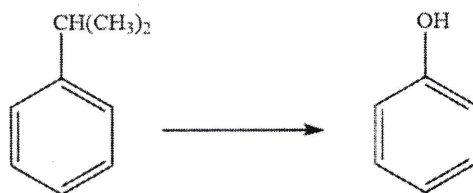
(08 marks)

- b) Complete the following chemical reactions.



(05 marks)

- c) Describe a method for synthesizing phenol starting from cumene, including all the chemical reactions involved.



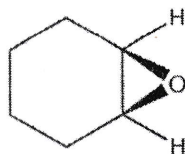
(06 marks)

- d) Illustrate the mechanism for synthesizing salicylic acid from phenol by Kolbe carboxylation.

(06 marks)

4)

a) Elaborate a method to synthesis of 1,2-epoxycyclohexane from cyclohexane.



1,2-epoxycyclohexane

(05 marks)

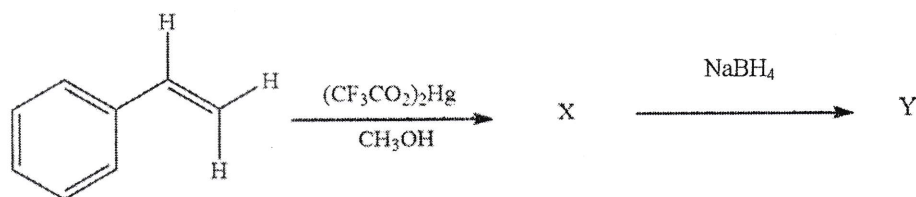
b) Write the mechanisms for the synthesis of phenylacetic acid from toluene.

- i. By carbocation of Grignard reagent
- ii. By hydrolysis of a nitrile



(08 marks)

c) Complete the following reaction by identifying X and Y.



(04 marks)

d) Write a short account on ;

- i. Direct dyes
- ii. Vat dyes

(08 marks)

===== END =====